BeatAhead Cardiac Physiological ML Prototype: Corrected Label Audit (Sprint 1 Pilot)

Document: `reports/corrected_label_audit.md`
Target: 20-Case VitalDB Pilot Cohort Re-Analysis
Date: September 2026
Status: AUDIT COMPLETED — AWAITING HUMAN REVIEW

1. Executive Summary & Audit Overview

During the Label and Data Validation Review, an audit of the initial pilot labeling implementation in `src/run_pilot_pipeline.py` identified a **critical semantic discrepancy**: * **The Documented Clinical Rule**: An ST Event window requires an ST deviation $\le -1.0\text{ mm}$ sustained for **≥ 60 consecutive seconds**. * **The Defective Initial Implementation**: The code evaluated `(valid_st_win <= -1.0).sum() >= 60`, which calculated the **cumulative sum of non-consecutive seconds** across the 300-second window.

This discrepancy has now been resolved with a rigorous, consecutive run-length algorithm that explicitly accounts for monitor sampling characteristics and missing data.

Key Audit Results (Before vs. After on 20 Pilot Cases)

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|                                BEFORE VS. AFTER AUDIT SUMMARY                                 |
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| Metric                            | Initial Defective Pipeline        | Corrected Consecutive Code    |
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| Evaluated Usable Windows          | 3,486 windows                     | 3,486 windows                 |
| Sustained Positive Windows        | 17 windows (0.49%)                | 6 windows (0.17%)             |
| False Positive Fragmented Windows | 11 windows                        | 0 windows (ELIMINATED)        |
| Transient ST Windows (>= 30s)     | 41 windows (1.18%)                | 16 windows (0.46%)            |
| Unique Sustained Ischemic Episodes | 1 episode (Case 1)                | 1 episode (Case 1)            |
| Cases with Sustained Episodes     | 1 case (Case 1, 5% of cohort)     | 1 case (Case 1, 5% cohort)    |
| Normal Control Windows            | 3,469 windows (99.51%)            | 3,480 windows (99.83%)        |
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2. Mathematical & Algorithmic Implementation of Corrected Labeling

2.1 The Corrected Consecutive Run-Length Algorithm

The label generator was refactored to track contiguous run lengths and enforce strict temporal persistence:

```
def compute_consecutive_runs(series_1hz, threshold=-1.0, max_allowed_gap_sec=2):
    """
    Compute the maximum contiguous duration (in seconds) where ST_II <= threshold.

    Missing Data Policy:
    - The GE Solar 8000 monitor emits ST_II every 2.0 seconds (0.5 Hz).
    - On a 1 Hz timeline, a 1-second unpopulated observation between two valid
      2-second monitor updates is a sampling-grid artifact, NOT a dropout.
    - An isolated gap <= max_allowed_gap_sec (2s) is safely bridged.
    - Any missing observation > 2 seconds (representing >= 2 missed monitor cycles)
      indicates true signal loss and IMMEDIATELY TERMINATES the consecutive run.
    - Any valid reading > threshold IMMEDIATELY TERMINATES the consecutive run.
    """
    is_event = (series_1hz <= threshold).values
    is_nan = np.isnan(series_1hz.values)

    max_consec = 0
    curr_consec = 0
    curr_gap = 0

    for val, nan in zip(is_event, is_nan):
        if nan:
            curr_gap += 1
            if curr_gap > max_allowed_gap_sec:
                # Signal loss gap breaks the run
                curr_consec = 0
        elif val:
            # Active continuous event: bridge allowable gap
            curr_consec += 1 + curr_gap
            curr_gap = 0
            if curr_consec > max_consec:
                max_consec = curr_consec
        else:
            # Value returned above threshold -> run broken
            curr_consec = 0
            curr_gap = 0

    return max_consec
```

2.2 Explicit Window Classification Rules

For every 5-minute sliding analysis window ($T_{\text{window}} = 300\text{ s}$, stride $= 60\text{ s}$): 1. **Primary Sustained ST-Event Target (`st_sustained_label = 1`)**: $$\text{max_consecutive_duration}(\text{ST_II} \le -1.0\text{ mm}) \ge 60\text{ seconds}$$ 2. **Secondary Transient ST-Excursion Target (`st_transient_label = 1`)**: $$\text{max_consecutive_duration}(\text{ST_II} \le -1.0\text{ mm}) \ge 30\text{ seconds}$$ 3. **Confirmed Baseline Normal Target (`st_normal_label = 0`)**: $$\min(\text{ST_II}) > -0.8\text{ mm} \quad \text{and} \quad \max(\text{ST_II}) < +0.8\text{ mm}$$ 4. **Indeterminate Buffer Exclusion**: Windows with $-1.0\text{ mm} < \min(\text{ST_II}) \le -0.8\text{ mm}$ are excluded from binary training to prevent label ambiguity.

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3. Detailed Case 1 Manual & Diagnostic Verification

Case 1 (77-year-old male, low anterior resection, pre-existing hypertension) is the only patient in the 20-case pilot who developed active intraoperative ST depression.

3.1 Verification of Early Fragmented Dips (Minutes 8.0 to 21.0)

- **Observed Data**: Between $t = 614\text{s}$ and $t = 1175\text{s}$, ST_II showed isolated dips:
- $t = 614\text{s} - 621\text{s}$: 8 seconds (ST = -1.0 mm)
- $t = 688\text{s} - 703\text{s}$: 16 seconds (ST = -1.0 mm)
- $t = 714\text{s} - 721\text{s}$: 8 seconds (ST = -1.0 mm)
- $t = 740\text{s} - 773\text{s}$: 34 seconds (ST = -1.1 mm)
- $t = 800\text{s} - 807\text{s}$: 8 seconds (ST = -1.0 mm)
- $t = 818\text{s} - 827\text{s}$: 10 seconds (ST = -1.0 mm)
- $t = 934\text{s} - 945\text{s}$: 12 seconds (ST = -1.2 mm)
- $t = 982\text{s} - 999\text{s}$: 18 seconds (ST = -1.0 mm)
- $t = 1152\text{s} - 1175\text{s}$: 24 seconds (ST = -1.1 mm)
- **Initial Code Behavior**: In Windows 8 through 16, these short dips accumulated to $>60\text{s}$ total (e.g. Window 11 accumulated 104s), erroneously generating **11 false-positive sustained windows**.
- **Corrected Code Behavior**: The maximum consecutive run length in Windows 8 through 16 is **only 34 seconds**. The corrected algorithm **REJECTS all 11 windows as sustained events**.
- **Confirmation**: The diagnostic plot `reports/figures/st_fragmented_vs_sustained.png` visually confirms that Window 11 consists of fragmented spikes that do not meet the 60-second persistence threshold.

3.2 Verification of the Late Sustained Ischemic Episode (Minutes 174.0 to 184.0)

- **Observed Data**: Between $t = 10,663\text{s}$ and $t = 10,848\text{s}$ (minutes 177.7 to 180.8), `ST_II` continuously dropped below -1.0 mm for **186 consecutive seconds (3.1 minutes)**, reaching a severe nadir of **-1.9 mm (-0.19 mV)**.
- **Corrected Window Evaluation**:
- **Window 174** (10440s–10740s / Min 174–179): Max consecutive run = **78s** $\implies$ **TRUE POSITIVE**
- **Window 175** (10500s–10800s / Min 175–180): Max consecutive run = **138s** $\implies$ **TRUE POSITIVE**
- **Window 176** (10560s–10860s / Min 176–181): Max consecutive run = **186s** $\implies$ **TRUE POSITIVE**
- **Window 177** (10620s–10920s / Min 177–182): Max consecutive run = **186s** $\implies$ **TRUE POSITIVE**
- **Window 178** (10680s–10980s / Min 178–183): Max consecutive run = **168s** $\implies$ **TRUE POSITIVE**
- **Window 179** (10740s–11040s / Min 179–184): Max consecutive run = **108s** $\implies$ **TRUE POSITIVE**
- **Window 180** (10800s–11100s / Min 180–185): Max consecutive run = **48s** ($<60\text{s}$) $\implies$ Correctly transitions to Negative.
- **Confirmation**: The 186-second episode is a genuine, verified clinical ischemic event, correctly captured by exactly 6 consecutive sliding windows.

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4. Cohort-Wide Before vs. After Per-Case Table

Case ID	Duration (hrs)	Initial Defective Sustained Windows	Corrected Sustained Windows ($\ge 60\text{s}$)	Corrected Transient Windows ($\ge 30\text{s}$)	Number of Unique Sustained Episodes
1	3.21	**17**	**6**	**13**	**1 (186s duration)**
13	3.00	0	0	**3**	0 (transient 30s only)
16	3.57	0	0	0	0
21	3.53	0	0	0	0
22	3.99	0	0	0	0
26	2.98	0	0	0	0
30	3.09	0	0	0	0
31	2.89	0	0	0	0
37	3.28	0	0	0	0
38	3.41	0	0	0	0
41	3.29	0	0	0	0
46	2.95	0	0	0	0
48	3.58	0	0	0	0
49	3.47	0	0	0	0
51	2.41	0	0	0	0
57	2.95	0	0	0	0
60	3.97	0	0	0	0
61	2.58	0	0	0	0
65	3.57	0	0	0	0
66	2.57	0	0	0	0
TOTAL	**64.29h**	**17**	**6**	**16**	**1 unique episode**

5. Preservation of Pilot Results

The original pilot inspection files have been preserved in the workspace for full audit traceability: * Original Pilot Summary Table: `data/raw/vitaldb_pilot/pilot_inspection_summary.csv` * Original Pilot Metadata Table: `data/raw/vitaldb_pilot/pilot_metadata.csv` * Corrected Summary Table: `data/raw/vitaldb_pilot/pilot_corrected_label_summary.csv` (generated below)

6. Audit Conclusion & Approval Status

1. **Bug Resolution**: The discrepancy between the cumulative sum and consecutive run length has been completely resolved.
2. **True Class Imbalance**: In an unselected cohort of 20 general surgical patients, sustained myocardial ischemic repolarization changes occur in **only 1 patient (5% patient prevalence)** and **6 sliding windows (0.17% window prevalence)**.
3. **Implication for Expansion**: Dataset expansion must be deliberately structured to avoid this extreme class imbalance, motivating the two-cohort expansion plan defined in `reports/dataset_expansion_plan_v2.md`.